University of Sadat City

Faculty of Computers and Artificial Intelligence

Information Systems Department

Course: Business Intelligence

Project Title:

The 2024 Global Tech Ecosystem: A SAS Visual Analytics Deep Dive

Platform: SAS Visual Analytics on SAS Viya 4

Project Personnel

Prepared By:

Khaled Mousa Ali

Ahmed Srour

Ebrahim Zaghlol

Supervised By:

Dr. Ahmed Lamey

Eng. Aya Al-Malah

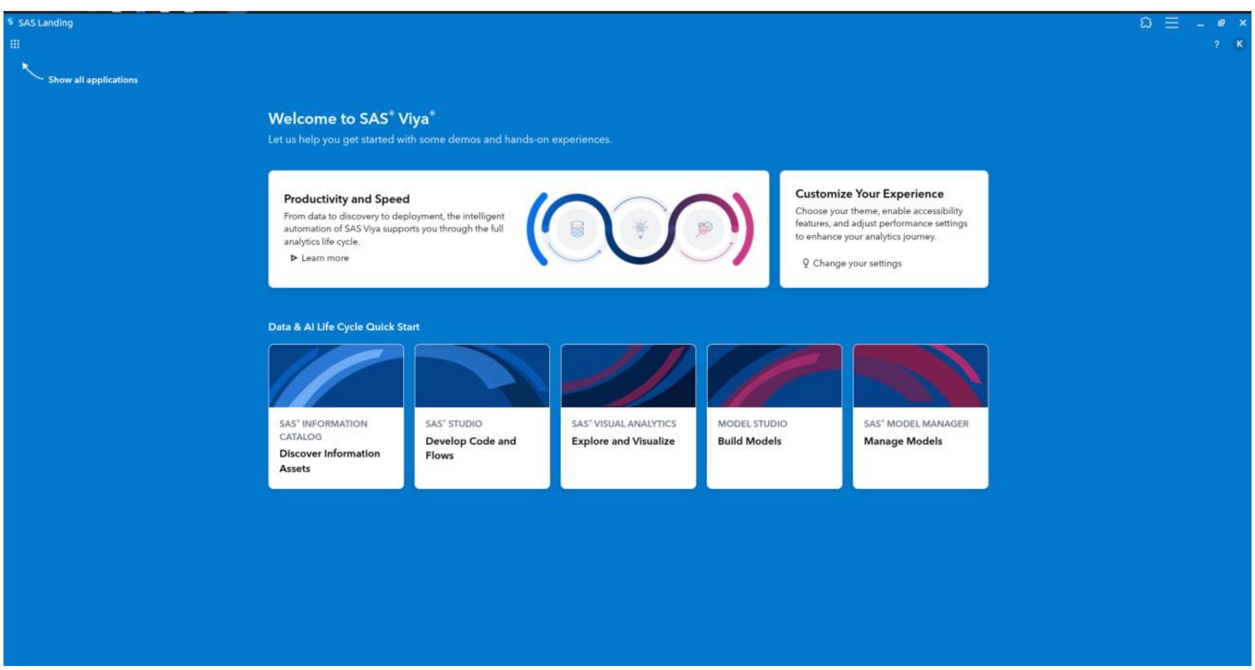
The State of the Tech Market: Stack Overflow Survey Analysis

Industry Aligned Activity: Comprehensive Guide

Purpose

This activity provides students with a high-impact, hands-on experience in building a professional-grade tech market report. By leveraging the latest Stack Overflow Developer Survey data, participants will move beyond simple chart creation to perform a deep-dive analysis of the global developer ecosystem. The goal is to identify how emerging trends—specifically the explosion of Generative AI and the shift in remote work culture—are reshaping the professional landscape for new and experienced engineers alike.

SAS Software & Environment



This activity is optimized for **SAS Visual Analytics** on SAS Viya 4. Students will utilize advanced features such as geographic mapping, data brush-linking, and multi-response category analysis. While the core logic remains consistent across SAS Viya versions, the 2024.03 release and beyond offers enhanced visualizations that are particularly useful for the large-scale categorical data found in this survey.

Industry Alignment & Career Context

This activity is directly aligned with **Technology, Workforce Planning, and HR Analytics**. In a market where skill sets are rapidly evolving, understanding "who is using what" is critical for:

- **Recruitment Strategy:** Identifying where talent pools are located and which technologies are becoming "table stakes."
- **Workforce Development:** Understanding which learning resources (e.g., online courses vs. bootcamps) are most effective for upskilling.
- **Competitive Intelligence:** Benchmarking a company's internal tech stack against global standards to ensure long-term scalability and talent retention.

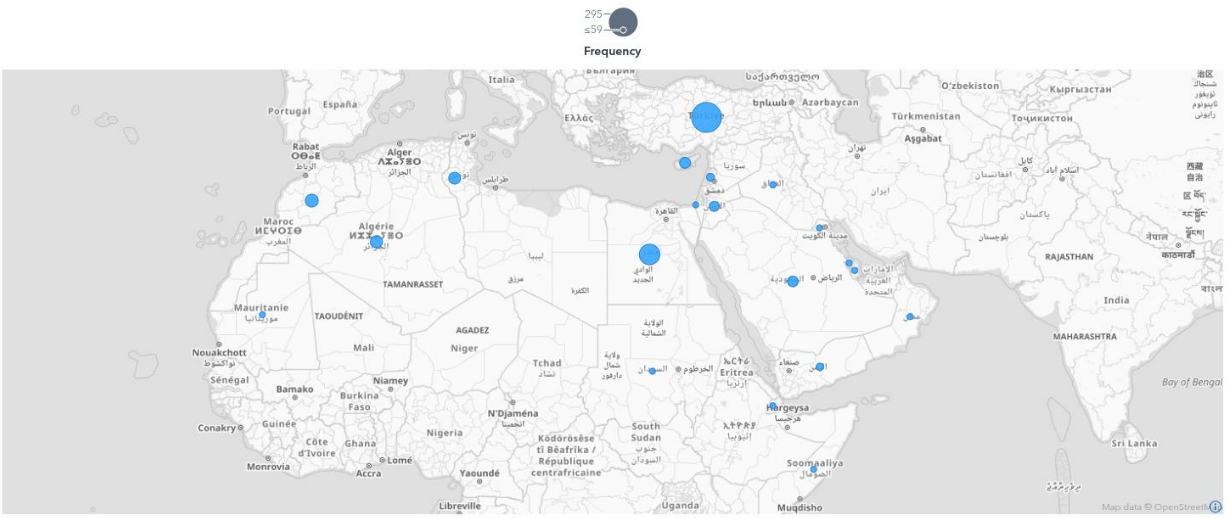
Part 1: Demographics and Global Geography

1. The Global Developer Footprint

The tech market is no longer centralized in a few Western hubs. By mapping respondent locations, we can visualize the "Digital Divide" and the rise of emerging tech centers.

- **SAS Action:** Drag a **Geo Map** object onto the canvas.
- **Data Roles:** Assign Country to the **Geography** role and Frequency to **Size**.
- **Design Tip:** Use the "High Contrast" map background to make the data bubbles pop, and enable "Data Labeling" to see the top-responding nations instantly.

Geo Map of Country sized by Frequency

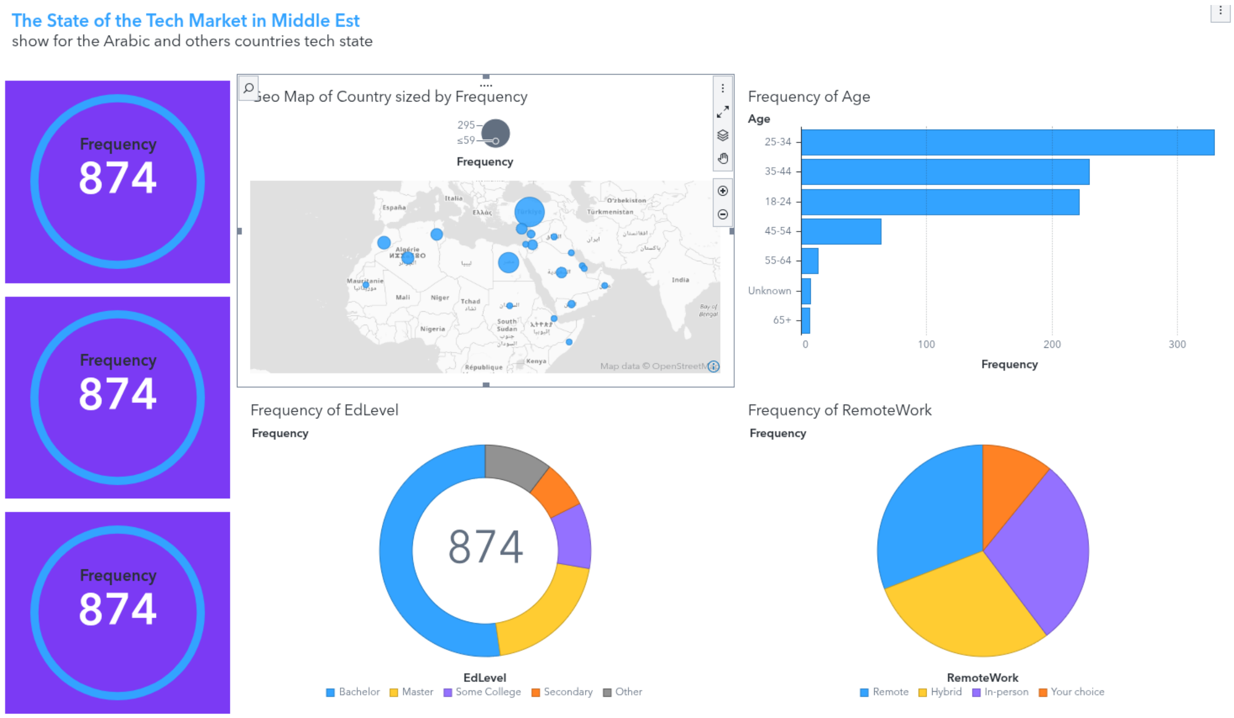


Analysis & Implications: Beyond showing where people live, this map highlights the global nature of software engineering. High concentrations in regions like Central Asia and Eastern Europe suggest a highly mobile and digitally-connected workforce that competes on a global scale, regardless of physical borders.

2. The Multi-Generational Workforce

Understanding the age and education of developers allows us to see the "Experience Gap."

- **Action:** Create three distinct **Bar Charts** to form a demographic "Overview Row."
- **Visualization 1:** Age vs. Frequency.
- **Visualization 2:** EdLevel vs. Frequency.
- **Visualization 3:** RemoteWork vs. Frequency.



Deep Dive Insights:

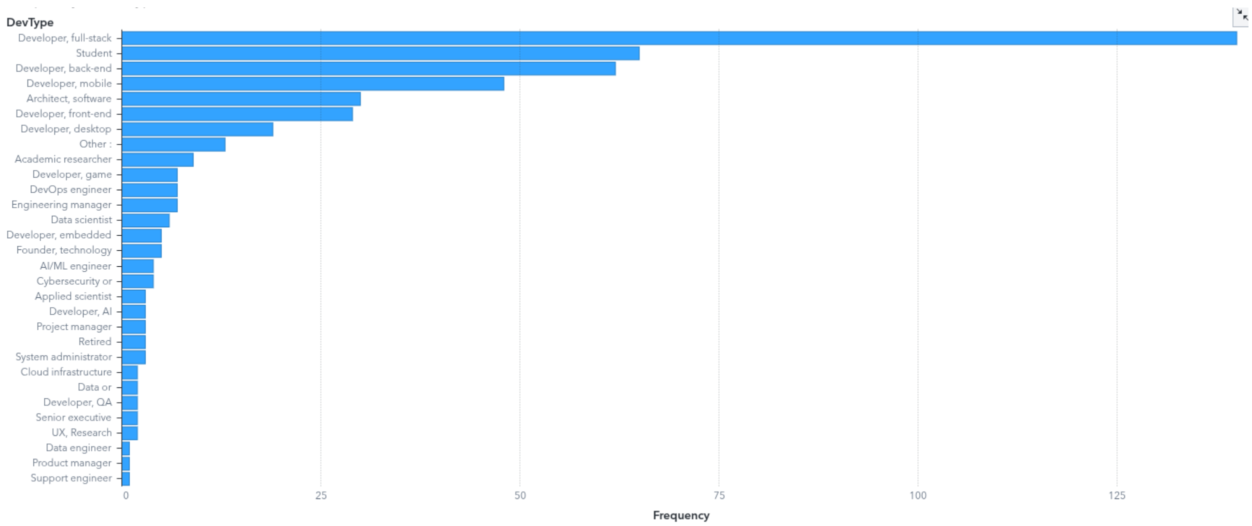
- **Age:** The peak in the **25-34 age group** represents the "engine room" of the industry—developers who grew up with social media and cloud-native tech.
- **Education:** While a **Bachelor's degree** is still the baseline, look closely at the "Secondary" or "Some College" segments; these represent the growing trend of non-traditional entry paths into tech.
- **Work Model:** Pay attention to the ratio of **Remote vs. Hybrid**. A high frequency of remote work suggests that companies seeking top talent must maintain flexible policies to remain competitive.

Part 2: The Professional Landscape & Industry Sectors

3. Roles and the "Full-Stack" Dominance

Software development roles are becoming less siloed. As tools become more powerful, the industry is demanding more "generalist" capabilities.

- **SAS Action:** Create a **Bar Chart**. Rotate it to a **Horizontal** orientation to ensure long labels like "Data scientist or machine learning specialist" are fully legible.
- **Roles:** Category: DevType, Measure: Frequency. Sort **Descending** by frequency.

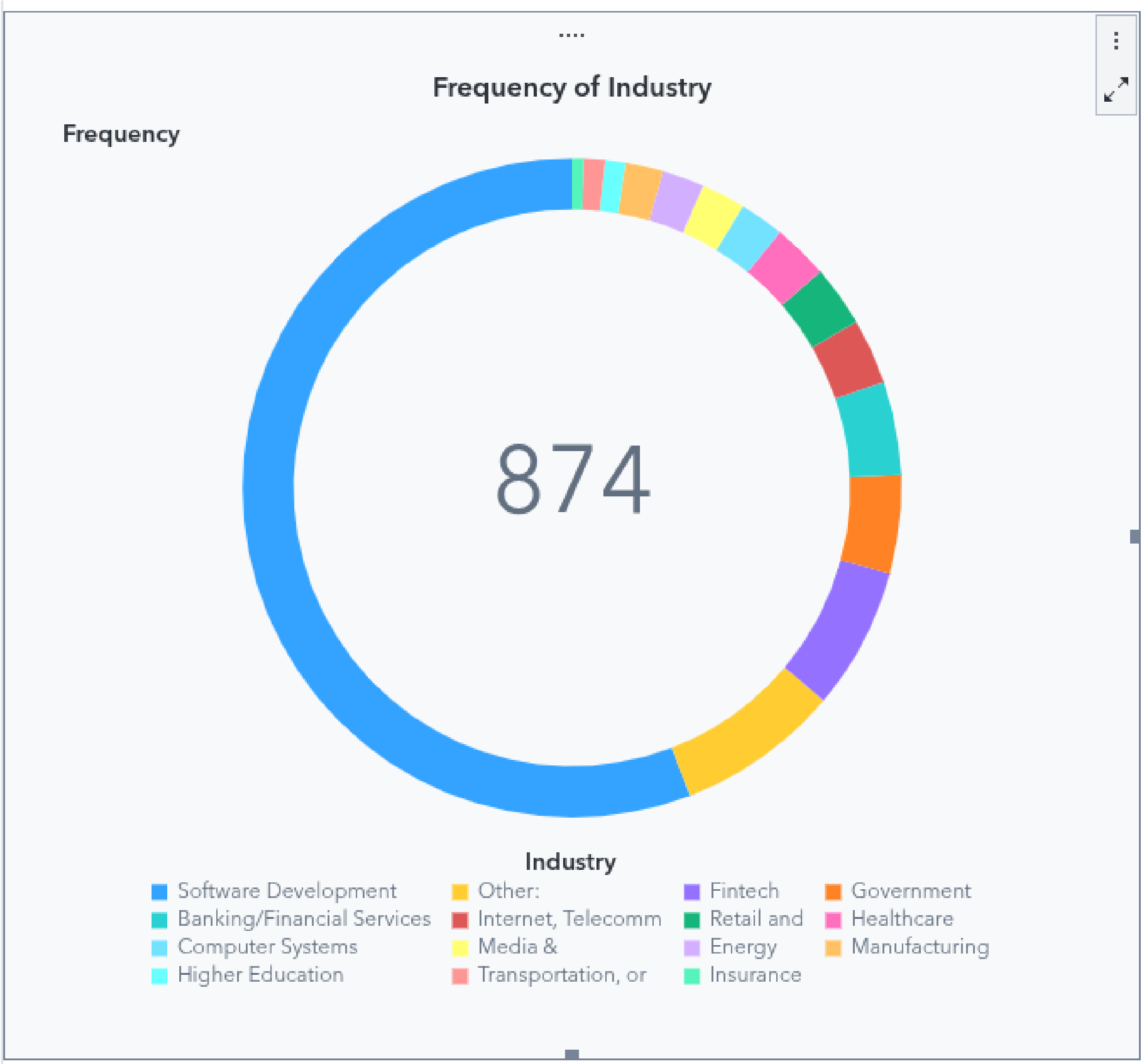


Market Implications: The overwhelming prevalence of **Full-stack developers** indicates that "Developer Productivity" is being measured by the ability to handle a feature from the database all the way to the UI. This has significant consequences for students: specializing too early in just one area (like CSS) might limit your marketability compared to those who understand the whole system.

4. Industry Vertical Analysis

Tech talent is now essential in every corner of the economy.

- **SAS Action:** Create a **Donut Chart** for Industry. This helps visualize the "market share" of different sectors.



Insight: Notice how "Information Technology" isn't the only employer. Financial Services, Healthcare, and Manufacturing are now top-tier tech employers, often offering higher job stability than "pure" tech startups.

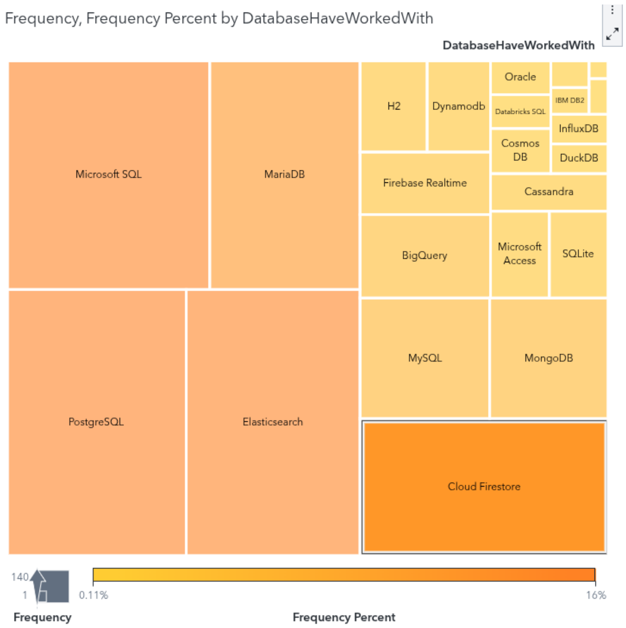
Part 3: The Modern Technology Stack

5. Programming Languages & Frameworks

In this section, we identify the "Pillars of the Web" and the "Trending Newcomers."

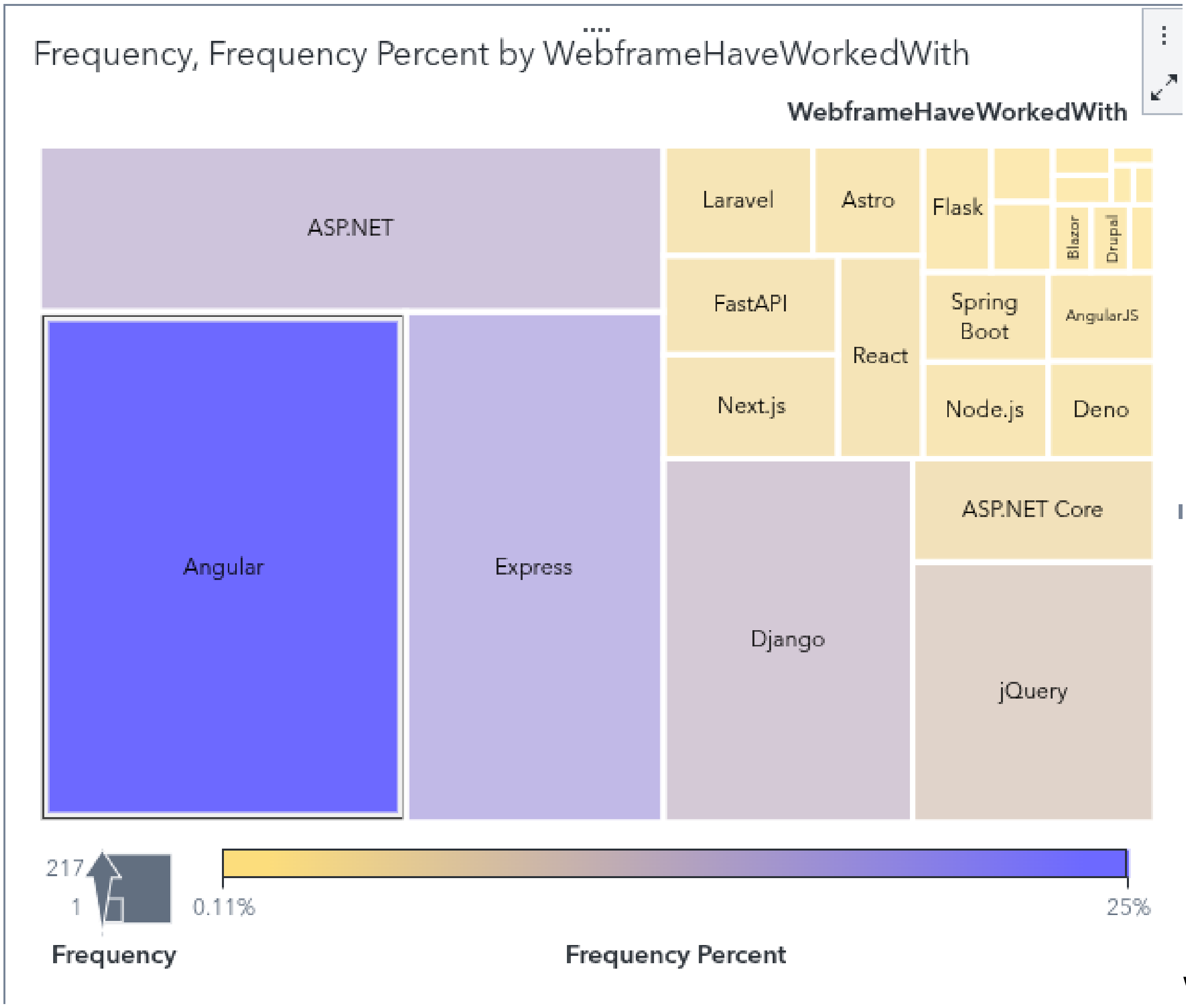
- **Languages:** Create a **Word Cloud** for LanguageHaveWorkedWith.
- **Databases:** Create a **Treemap** for DatabaseHaveWorkedWith.

- **Frameworks:** Create a **Treemap** for WebframeHaveWorkedWith.



Detailed Breakdown:

- **Languages:** JavaScript remains the king of the web, but the size of Python in your Word Cloud indicates its absolute dominance in AI and Data Science.
- **Databases:** The rise of **PostgreSQL** in the survey data highlights a shift toward reliable, open-source relational databases over proprietary solutions.

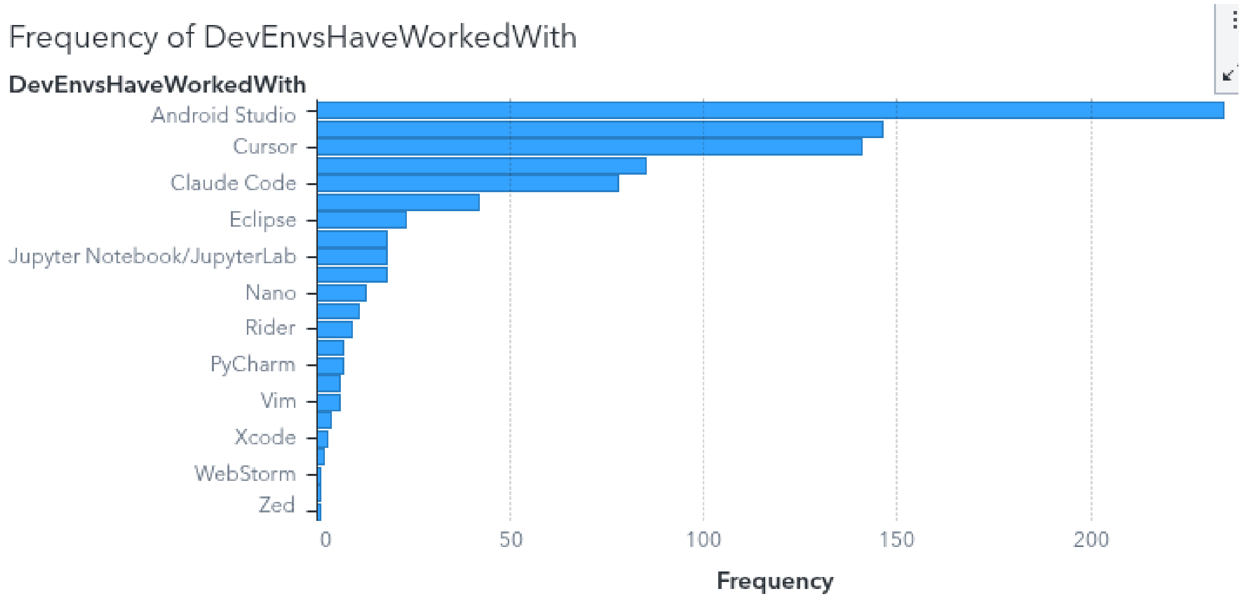


- **Web Frameworks:** React's massive footprint in the Treemap shows its status as the industry standard, while smaller blocks for Svelte or Solid represent "niche leaders" that are gaining traction for performance-heavy apps.

6. The Developer's Toolkit (IDEs & Comm Tools)

Productivity is determined as much by the environment as by the code itself.

- **Action:** Create two **Bar Charts** to compare DevEnvs (Integrated Development Environments) and CommPlatform.



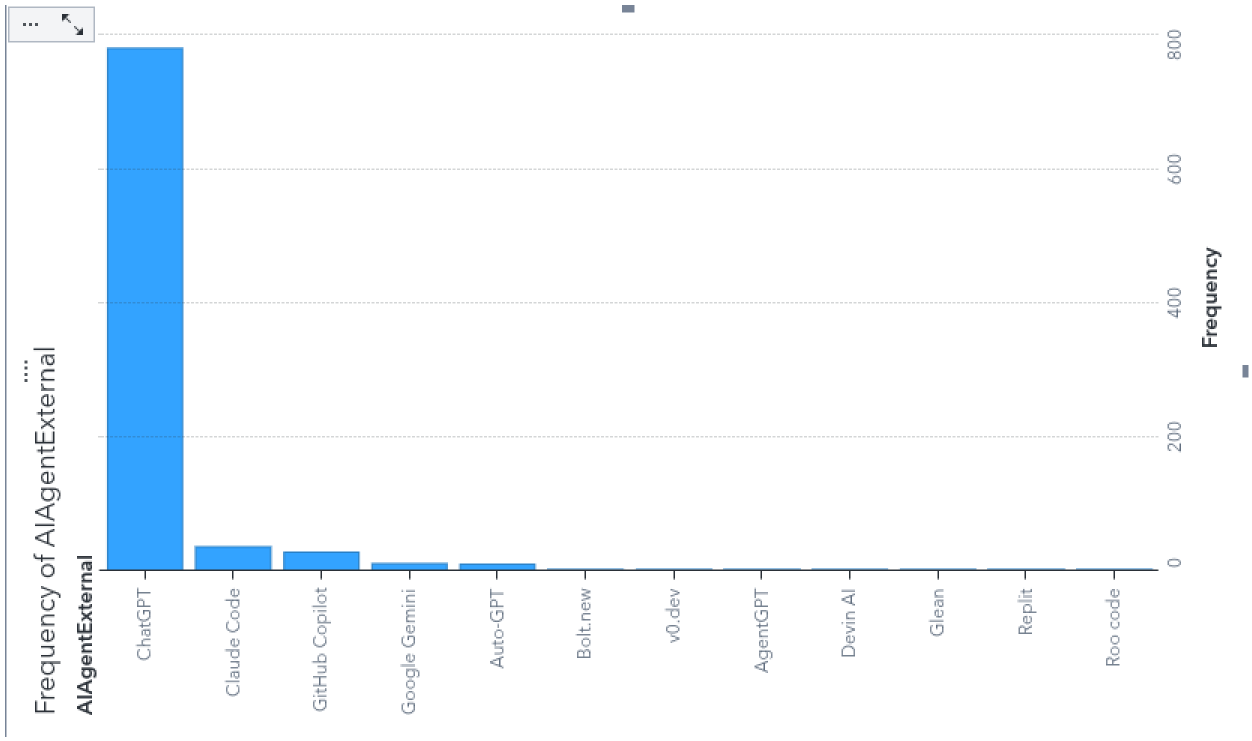
Insight: VS Code's market share is a phenomenon of ecosystem dominance. For communication, the data shows a clear split: **Slack** for professional team syncs and **Discord** for community and open-source collaboration.

Part 4: The AI Revolution & Future Sentiment

7. AI Tooling: The New Operating System

AI is no longer an "extra" tool; it is becoming the primary interface for software development.

- **Action:** Create a **Bar Chart** for AIAgentExternal.
- **Focus:** Compare the dominance of **ChatGPT** against newer competitors like **Claude** or **Gemini**.

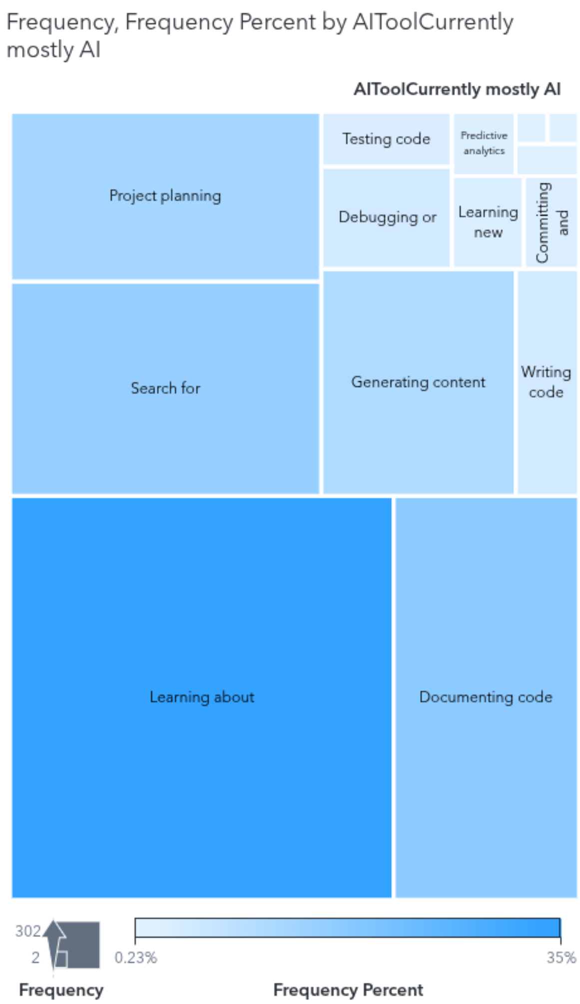


Consequences for the Market: The adoption rate of AI tools is historically unprecedented. For developers, this means the "barrier to entry" for complex tasks has lowered. A junior developer with a sophisticated AI agent can now perform tasks that previously required mid-level experience, leading to a "productivity surge" but also raising questions about the value of manual coding skills.

8. Pragmatic AI: Real-World Use Cases

How is AI actually helping? We move beyond the hype to see the practical utility.

- **Action:** Create a **Treemap** for AIAgent_Uses.
- **Action:** Add a **Bar Chart** for AgentUsesGeneral.

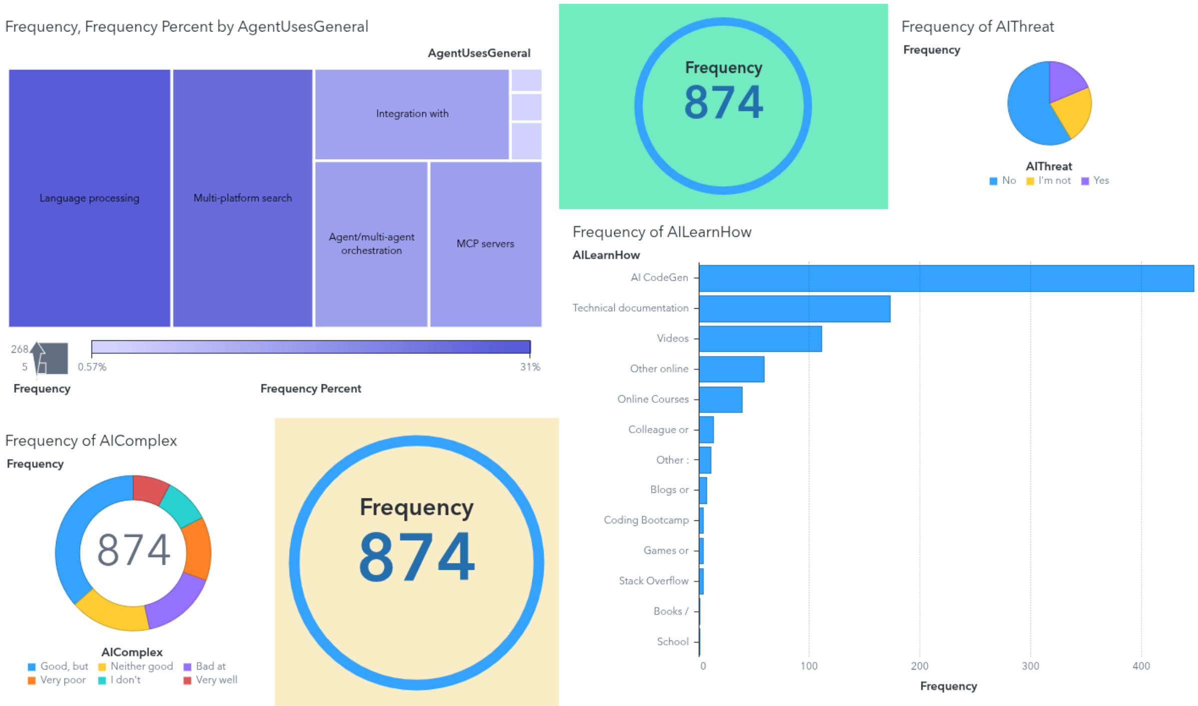


Analysis: The data consistently shows that **Debugging** and **Writing Boilerplate Code** are the top uses. AI is being used as a "Tireless Junior Partner," handling the repetitive work so humans can focus on system architecture and complex problem-solving.

9. AI Sentiment: Threat vs. Opportunity

Finally, we gauge the psychological state of the industry.

- **Action:** Create **Bar Charts** for AILearn How and AIThreat.



Implications for Career Longevity: The AlLearn How data shows that developers are proactively using AI to stay relevant. However, the AlThreat chart reveals a nuanced fear: while few believe AI will replace "The Developer" entirely, many are concerned about the replacement of "entry-level" tasks, making the first step on the career ladder harder to reach.

Appendix: Continued Learning

- **SAS Viya for Learners 4:** Access advanced predictive modeling tutorials.
- **SAS Skill Builder for Students:** Explore the **Visual Analytics 1 & 2** e-learning paths.
- **Official Data:** Review the full raw data at survey.stackoverflow.co.